

Saiyma Sittul Muna

Prospective Ph.D. Student

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 Muna-sit
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 Sittul Muna

Research Interests

Large Language Models (LLMs), Natural Language Processing (NLP), Mechanistic Interpretability, AI Agents and Agentic Systems, Uncertainty Quantification, Efficient LLMs (Quantization and Parameter-Efficient Fine-Tuning), Retrieval-Augmented Generation (RAG)

Education

Islamic University of Technology (IUT)
B.Sc. in Computer Science and Engineering
CGPA: 3.65 / 4.00

Jun 2021 – Oct 2025
Gazipur, Bangladesh

Academia School
O and A Levels

2018 – 2020
Dhaka, Bangladesh

Summerfield International School
High School

2009 – 2017
Dhaka, Bangladesh

Research Works and Publications

“Prompting with Sign Parameters for Low-resource Sign Language Instruction Generation”
Accepted to: ICCV 2025 Workshop on Computer Vision for Accessibility, July 2025

- Introduced **BdSLIG**, the first benchmark for Bengali Sign Language Instruction Generation (SLIG), and proposed **Sign Parameter-Infused (SPI) prompting**, a structured prompting framework that incorporates linguistic sign parameters into Vision-Language Model reasoning.

“FirstAidQA: A Synthetic Dataset for First Aid and Emergency Response in Low-Connectivity Settings”

Accepted to: 5th Muslims in ML Workshop co-located with NeurIPS 2025

- Introduced **FirstAidQA**, the first synthetic benchmark for first-aid instruction generation, expert-validated question-answer pairs. Developed an **LLM-driven synthetic data generation and validation framework** for constructing safety-critical instruction-tuning corpora.

“Evaluating Retrieval-Augmented Generation Variants for Biomedical Multiple-Choice Question Answering”

Accepted to: IEEE International Conference on Future Machine Learning and Data Science 2026

- Developed **BanglaMedQA** and **BanglaMMedBench**, the first large-scale Bangla biomedical MCQ benchmarks, and constructed an OCR-derived biomedical retrieval corpus. Designed and systematically evaluated **multiple Retrieval-Augmented Generation (RAG)** framework encompassing adaptive retrieval routing, iterative feedback, evidence aggregation, and agent-driven retrieval policies.

“HandsOnAid: Building and Evaluating an Offline First-Aid Assistant When No Help is Coming”

Submitted to: 6th ML4H Symposium (Machine Learning for Health)

- Developed an **offline first-aid LLM** by fine-tuning **Gemma 2B with QLoRA** for low-connectivity emergencies. Explored medical LLM safety in no-EMS settings. Built a blinded, per-item multi-judge evaluation framework to examine fine-tuning, RAG, quantization, and self-critique, along with on-device optimization and corpus-level failure analysis.

“Uncertainty-Guided Error Repair in Multi-Step Language Model Agents: When to Target, When to Restart”

Submitted to: AAAI Conference on Artificial Intelligence (AAA - 27)

- Derived **step-level uncertainty profiles** for failed ReAct trajectories from token-, sampling-, and self-report-based signals, using standard and cascade-aware localization rules to identify repair origins. Evaluated prefix-preserving suffix regeneration with backtracking against judge-targeted, full-restart, and random-step baselines.

Ongoing Research Projects

- **Building Fairness-Aware Quantized Large Language Models** Investigating the **mechanistic and explainability aspects of neural network** quantization to understand how quantization modifies internal computations that influence fairness. Analyzing benchmark quantization methods to identify bias-inducing mechanisms and leveraging these insights to develop a **fairness-aware quantized LLM**.

Work Experience

Research Assistant

ELITE Research Lab LLC

June 2026 – Present

New York, USA

- Conduct research on multimodal representation learning, uncertainty-aware learning, probabilistic inference, and autonomous agentic systems for trustworthy AI.
- Design and implement large-scale experimental pipelines for calibration, uncertainty propagation, robustness evaluation under distribution shift, and reproducible model assessment.

Software Engineer

Technovative Solutions Limited

July 2024 – Oct 2024

Dhaka, Bangladesh

- Worked with AI technologies, including LLMs and RAG modeling
- Conducted thorough research on the given topic to ensure accuracy and depth of information

Internship

Teletalk

Jun 2024 – July 2024

Dhaka, Bangladesh

- Managed and streamlined billing processes for customer accounts
- Provided IT support and troubleshooting for internal software and hardware issues

Part-time Teacher

ALOHA

Feb 2018 – Jun 2018

Dhaka, Bangladesh

- Used ALOHA teaching methodologies to deliver lessons effectively
- Ensured a conducive learning atmosphere

Projects

- MoodDecoder** 2025
- Speech Emotion Recognition system comparing Wav2Vec2.0 and HuBERT on the MELD dataset with a real-time emotion detection interface built using Gradio.
- DigitAid** 2024
- DigitAid is a specialized app employing machine learning to recognize handwritten digits by dyslexic individuals.
- Safety Sync App** 2023
- Flutter-based mobile application featuring live location sharing, friend interactions, self-defense tutorials, mood-based meditation, and user feedback.
- Shape Shift Website** 2023
- Health and wellness platform offering workout tutorials, meal plans, and meditation resources for healthy living.
- Rapid Red App** 2023
- Java application that connects emergency patients with nearby blood donors to reduce response time during critical situations.

Technical Skills

Programming Languages: Python, Java, C++, Dart, JavaScript, HTML/CSS, LaTeX

ML & Deep Learning: PyTorch, Hugging Face Transformers, PEFT/QLoRA, model quantization, RAG pipelines, Gradio

Scientific Computing & Data: NumPy, Pandas, SciPy, Statsmodels, MATLAB

HPC & Developer Tools: Linux, SLURM, CUDA, GPU acceleration, Git, Jupyter, Google Colab

Databases: MySQL, MongoDB, Firebase, Oracle

Spoken Languages: Bengali, English, Hindi, Spanish

Extracurriculars

- Mentee at Banglalink Womentor 6.0 (a program for women in STEM)
- 60% Scholarship in ISCEA PTAK Prize Case competition 2023
- 2nd runners-up in AIUB presented idea contest on 'Introduction to Robotics.'
- 2nd runners-up in Aloha International Mental Arithmetic Competition 2012

Test Scores

IELTS : Overall: 8.0; Listening: 8.5; Reading: 8.0; Speaking: 7.5; Writing: 7.0